LINUX INSTALLATION

LINUX DISTRIBUTIONS

- Recall: What's a Linux Distribution?

- These are common across all distributions
  - The Linux Kernel (the core OS component)
  - The default GNU software tools (e.g. ls, rm, etc)
  - Some general software expected of a Linux distro (e.g. vi)

WHERE DO LINUX DISTROS DIFFER?

- Hardware Compatibility
- Bundled Software
- Community Support
- Desktop Environment
- Stability
- Cost

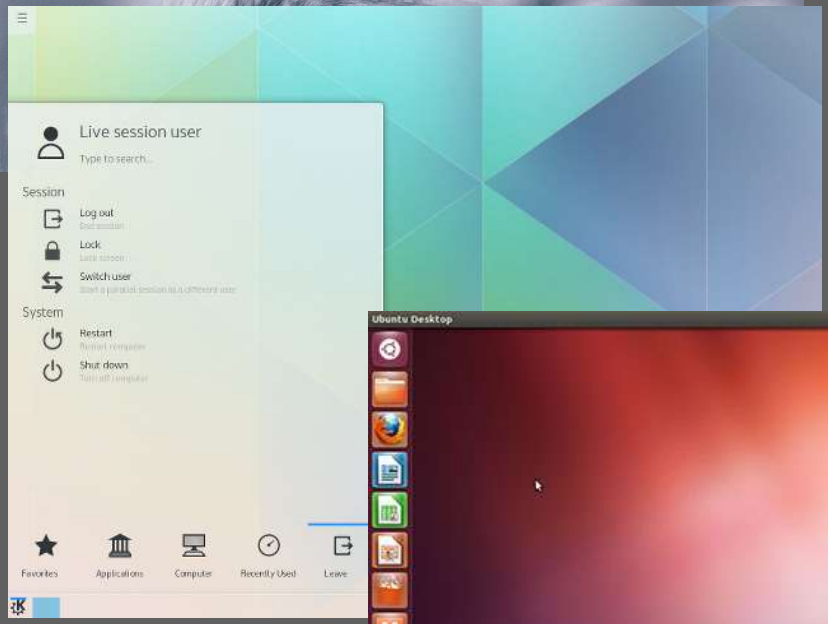
DESKTOP ENVIRONMENTS

- How do they differ?
  - The basic look and feel
  - Customizability
  - Resource usage
  - GUI options
  - Program integration with desktop

DESKTOP ENVIRONMENTS

GNOME

KDE



UNITY

STABILITY

Stable

- Longer update cycles
- Generally less buggy
- E.g. Debian Linux

Cutting Edge

- Software always up to date
- Risk of buggy software
- e.g. Fedora Linux

COST - I THOUGHT LINUX WAS FREE…?

- Paid versions of Linux offer some extras that are not available on free distros.
  - Printed manuals
  - Installer disks
  - Vendor support for a particular period of time
  - Guaranteed service level response (e.g. enterprise editions)
  - Sometimes you may get more software than with other distributions (eg extra DVDs instead of downloads).
  - Commercial software titles (e.g. copyrighted / patented technologies)

CHOOSING A DISTRO

Distribution	Cost	Suitable For	Notable Features
RHEL	paid	Server deployments	Telephone based support, stability
OpenSUSE	Free	Beginner to Advanced	Good documentation and support
Arch Linux	free	Advanced users	Highly customizable
Ubuntu	free	Beginner to advanced users and Server deployments	Own desktop environment, large software selection
Linux Mint	free	Beginners	Easy to use, comes with commonly used proprietary software and drivers
Gentoo	free	Advanced Users	Majority of software is compile-on-your-own
Fedora	free	Intermediate to advanced	Short update cycle (6 months)
Debian	free	Intermediate to advanced	Highly stable, long update cycles (1-3 years)

CHOOSING A DISTRO

In general, choose the Linux distribution that is closest to your ideal setup.

INSTALLATION TASKS

- Linux installation procedures usually need the following basic tasks:

Set the time zone.

Setting up the disk partitions

Create, user account and password

Select / install software packages

LINUX DISK PARTITIONS

- Linux requires at least one partition, for the *root filesystem* to hold the Linux kernel
- Optional
  - A *swap partition* can be created to be used as virtual memory
  - Separate partitions to hold other system components (e.g. \usr for software, \home for personal files
- Why have multiple partitions
  - Usually makes the system boot up faster
  - Less tedious to backup and restore in case of issues
  - Prevents files from growing too large

INSTALLING FROM SOURCE CODE

- Sometimes, software needs to be compiled prior to installation
  - Pre-built package is not available
  - Need certain software options that are not enabled in the pre-built package
- To get software in source form
  - Download from developer repository; or
  - Download using the package manager but specify to get the source instead of a precompiled version (e.g. apt-get build-dep)

BUILDING SOFTWARE

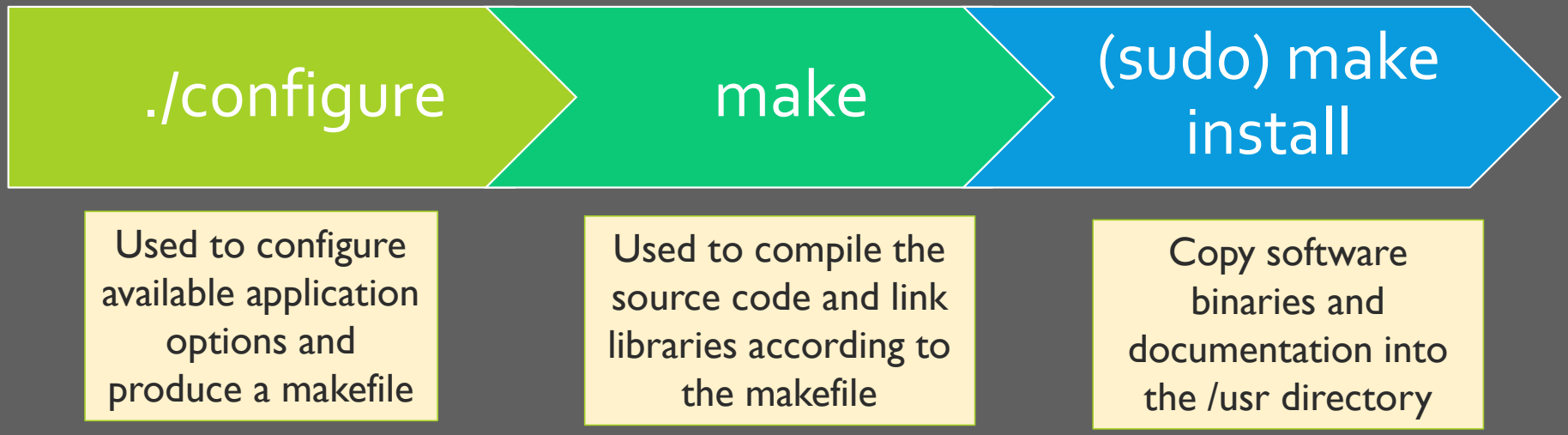
- Downloaded packages are usually in compressed archives (tarballs)
- Extract using the command

```
tar <options> <filename>
```

Options	Use for
x	Extraction
c	Creation
z	For files with extension .tgz or .tar.gz
j	For file with extension .tbz, .tbz2, .tar.bz2, or .tar.bz
v	verbose
f	Following is archive name

BUILDING SOFTWARE

- GNU Autotools - aka GNU build system
  - A suite of tools that help simplify the building of programs distributed as source code across different platforms



MANAGING SOFTWARE PACKAGES

- At installation, you will usually have the option to include some available packages in addition to kernel installation
- To add more packages in the future, you can use a Package Manager – collection of tools that automates the installation, configuration and removal of system software
- Usual capabilities
  - Software lookup from a repository
  - Software download, extraction and installation in standard locations
  - Resolving software dependencies
  - Version tracking and update

MANAGING SOFTWARE PACKAGES

- Debian Family (includes Ubuntu)
  - Software packages are distributed as files with a '.deb' extension
  - Package Managers
    - dpkg - lowest level package manager
    - Advanced Package Manager (APT) – Command line front-end for dpkg to make it a little more 'user friendly'
- Red-Hat Family (CentOS, Red Hat, Fedora)
  - Software packages are distributed as files with a '.rpm' extension
  - Package Managers
    - Rpm – low level command line
    - Yum – Command line front end

MANAGING SOFTWARE PACKAGES

- Dpkg
  - dpkg –l - lists installed packages and their versions
  - dpkg –L <package> - lists files associated with a package
- Advanced Package Manager (APT)
  - apt-get update - Reads the /etc/apt/sources.list file and updates the system's database of packages available for installation
  - apt-cache search - Looks for packages with keywords that can match the given string
  - apt-get install - Installs the package(s) specified, along with any dependencies
  - apt-get remove - Removes the package(s) specified, but does not remove dependencies and configurations
  - apt-get --purge remove- Removes package(s) and downloaded files including configuration
  - apt-get upgrade - Upgrades all packages if there are updates available (after an apt-get update)

SHUTTING DOWN

shutdown [time] [message]

- Shuts down or reboots the system

Option	What it does
-h	Turns off the system (alternative command: poweroff)
-r	Reboot the system (alternative command: reboot)

- Time may be specified as
  - hh:mm – specifies time to shutdown
  - +m – specifies number of minutes from current time
  - Now – immediate shutdown